

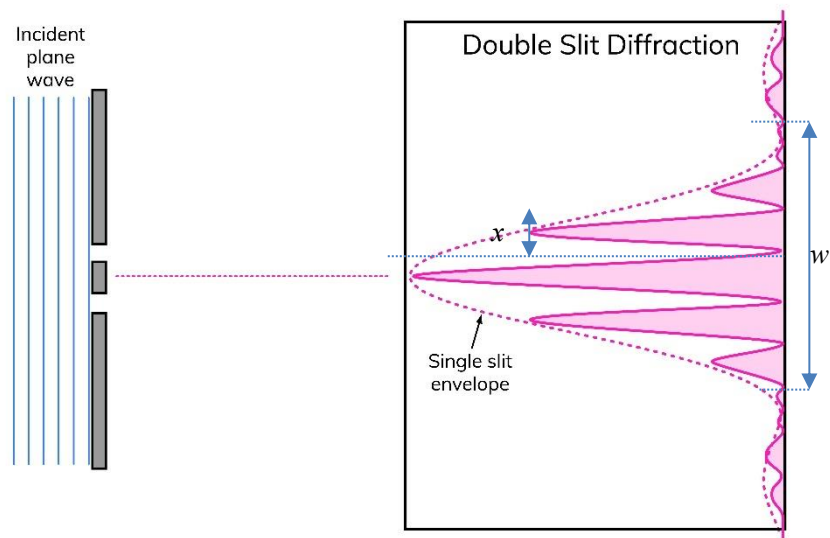
7(c)(ii)1.	Destructive interference occurs when light from second slit meets anti-phase with light from first slit.	B1
CKW MC	This part was well done	
7(c)(ii)2.	$x = \frac{\lambda D}{a} = \frac{(633 \times 10^{-9})(5.12)}{0.240 \times 10^{-3}}$ $= 0.0135 \text{ m}$	C1 A1
CKW MC	This part was well done	
7(c)(ii)3	$w = 6x \approx 0.0810 \text{ m}$ (The 3rd maxima from the central maxima of the double-slit interference pattern coincides with the 1st order minima of the single slit spectrum due to the width of the slit.) These places are minima of the single slit envelope.	B1 B1

CKW
MC

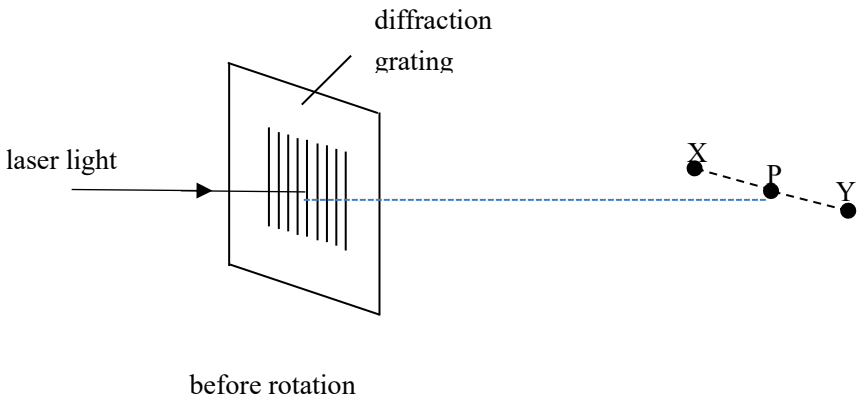
Realistic Double Slits

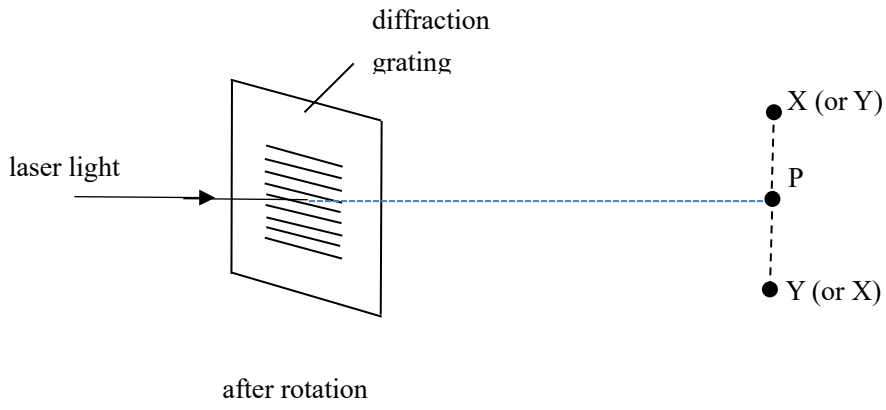
In real life, the double slits are never infinitely thin (they are not really point sources), they actually have a width.

- This means that each of the two slits will generate a single-slit diffraction pattern.
- The result is that we have a **single-slit envelope** with the pattern of dark and bright fringes from the actual double slit interference.



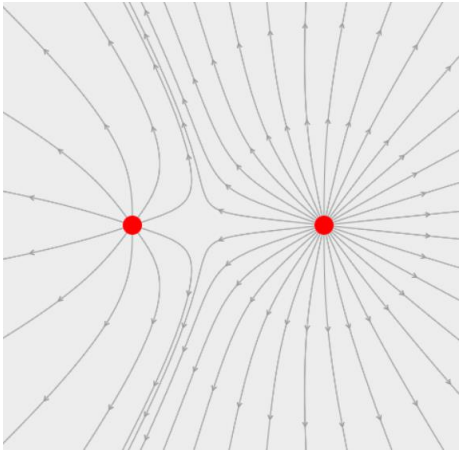
- To get information about the **distance** between the two slits, you have to look at maxima generated by the **double slit pattern**.
- To get information about the **width** of each slit, you have to look at the **position of the 1st order minima** of the single slit envelope.

7(d)(i)	$\tan \theta = \frac{38}{165} = 0.230$ or $\sin \theta = \frac{38}{\sqrt{38^2 + 165^2}} = 0.224$ or $\theta = 12.97^\circ$ $d \sin \theta = \lambda$ $d \sin 13^\circ = 6.33 \times 10^{-7}$ $d = 2.82 \times 10^{-6} \text{ m}$ number per metre = $\frac{1}{d}$ = $3.5 \times 10^5 \text{ m}^{-1}$	C1 C1 A1
CKW MC	- Most candidates omitted the unit. - Small angle approximation is not needed. $\theta = \frac{s}{r} = \frac{38}{165} = 0.230 = \tan \theta \neq \sin \theta$	
7(d)(ii)	1 Lines are further apart in second order, than in first order. 2 Lines are brighter and sharper in first order, than in second order.	B1 B1
CKW MC	- Most candidates did not provide a complete comparison of the shape of the maxima i.e. both the intensity and width. - The wavelengths of 633 nm and 638 nm are too close for the colours to be different.	
7(e)(i)	P remains in the same position. X and Y rotate through 90° .	B1 B1
 before rotation		

 after rotation		
CKW MC	- It is not possible to deduce if the rotation is clockwise or anticlockwise. - It is important to identify that the position of P is not affected, as it is possible that there is a translation in the diffraction pattern.	
7(e)(ii)	screen is not parallel to grating OR grating is not normal to incident light	B1
CKW MC	Many candidates did not realise the same grating was used, so the adjacent slit centre-to-centre separation remains unchanged.	

8(d)(i)	For gravitational potential at a point, the displacement of a small test mass from infinity to that point is always in the opposite direction to the external force acting on the small test mass because gravitational forces between two masses are always attractive. For electric potential at a point, the displacement of the small test charge from infinity to that point can be in the opposite or same direction to the external force acting on the small positive test charge because electric forces between two charges can be attractive or repulsive.	B1 B1
	OR [w.r.t. to DHS G.Field Tut] Due to the attractive nature of the gravitational force, work done by an external force to bring any mass from infinity to that point is always negative. However, electric force can be attractive or repulsive, thus work done by an external force to bring any charge from infinity to that point can be positive or negative.	B1 B1
CKW MC	Some candidates referenced the formulae for gravitational and electric potential in their discussions but failed to address the key conceptual difference between the two types of potentials, despite their similar definitions i.e. in terms of work done by an external force.	
8(d)(ii)	The electric force between an alpha particle and a gold nucleus is many orders of magnitude larger than the gravitational force between the alpha particle and gold nucleus.	B1

CKW MC	- A correct identification of the forces involved, and their comparison is needed to successfully answer this part. - No matter how small the gravitational force between an alpha particle and a gold nucleus is, if it is the only force present, then it cannot be neglected. - The order of magnitude of a gravitational force between two point masses is not only dependent on the product of the masses but also on the gravitational constant and the separation of the masses. - At the same separation r, the electric force is given by $F_e = \frac{(79e)(2e)}{4\pi\epsilon_0 r^2}$ while the gravitational force is given by $F_g = G \frac{(197u)(4u)}{r^2}$. Thus, $\frac{F_e}{F_g} = \frac{(79e)(2e)}{4\pi\epsilon_0 G(197u)(4u)}$$= \frac{(79)(2)(1.6 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(6.67 \times 10^{-11})(197)(4)(1.66 \times 10^{-27})^2}$$\approx 3 \times 10^{35}$	
8(d)(iii)	Assumption: The gold nucleus acts as a point positive charge. $V = \frac{Q}{4\pi\epsilon_0 r}$ $= \frac{79(1.6 \times 10^{-19})}{4\pi(8.85 \times 10^{-12})(2.6 \times 10^{-12})}$ $= 43714$ $= 44000 \text{ V}$	B1 C1 A1
CKW MC	- The assumption that “the gold nucleus is isolated” is not valid since the electric potential required is specified to be due to the gold nucleus in the question. - The formula used is only applicable for a point charge, which the gold nucleus is not physically in reality. - Some students thought the charge of a proton is 1 C.	

8(d)(iv)		B1: Direction of arrows B1: Position of neutral point (closer to alpha particle) B1: Field line pattern: • Field lines do not intersect. • Density of field lines around alpha particle is less than gold nucleus. • Symmetrical about the horizontal, asymmetrical about the vertical. • The circle represents an equipotential volume. The field lines are normal to the surface. [The suggested answer shows the simulated field lines between a charge of +1 C and +5 C. The actual field lines between a charge of +2e & +79e is more asymmetric.]
CKW MC	• Some candidates sketched ○ equipotential lines, or ○ magnetic field lines • instead of electric field lines. • The imaginary line joining the points of zero resultant electric field strength were observed to vertically straight in many cases; this is only possible if the charges have the same magnitude and same sign. Some sites to simulate electric field lines between two point charges: ○ E-Field Simulation - Ithaca College PER ○ Electric field line simulator	